Yi Xia (Sean)

Applied AI & ML Software Engineer

Open to: Applied AI Engineer; Machine Learning Engineer; Computer Vision Engineer; Data Scientist / Applied Scientist; Software Engineer (AI/ML)

Auckland, New Zealand · +64 27 368 7961 · xiayi.shawn.001@gmail.com

LinkedIn: linkedin.com/in/yi-xia-571a83267 · Google Scholar: scholar.google.com/citations?user=CEyoI9IAAAAJ

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SUMMARY

Auckland-based applied AI engineer and PhD candidate who builds production computer-vision systems. Four years at TOHU MEDIA designing and deploying object detection, OCR and action-recognition pipelines on real film-industry video, with PyTorch training, Docker-packaged inference on cloud GPU servers, and direct iteration with non-technical users. Concurrently completing an AUT PhD in deep learning for human activity recognition; two first-author Q1 journal papers (Pattern Recognition and Computer Science Review), seven peer-reviewed papers and 70+ citations. Strong fit for New Zealand teams needing practical AI, ML engineering, video analytics, document/OCR AI, agritech, healthcare, insurance, transport or enterprise data-science work.

TECHNICAL SKILLS

Programming and ML: Python, Java, PyTorch, TensorFlow/Keras, scikit-learn, NumPy, pandas, Hugging Face.

Computer vision: YOLO family, object detection, object tracking, OCR, video analytics, action recognition, segmentation, classification.

Deployment: Docker, containerised inference, Linux, Git/GitHub, CUDA, cloud GPU servers, model evaluation.

Research methods: GCNs, CNNs, attention/Transformers, skeleton-based HAR, frequency-domain learning.

Generative AI: Claude/GPT API use, LLM integration, agentic workflows, function/tool calling, prompt engineering.

EXPERIENCE

Artificial Intelligence Developer — **Feb 2022 - present**

TOHU MEDIA, Auckland - full-time since June 2025; part-time alongside master's and PhD study before that

- Own the company's AI capability end-to-end: scoped datasets, trained/evaluated models in PyTorch, packaged inference in Docker, and deployed to company cloud GPU servers.
- Designed and shipped three production AI systems - object detection, OCR and action recognition - replacing previously manual transcription work on client video footage (specific deliverables under NDA).
- Built for messy real footage: variable cinematography, dense overlay text, low-contrast archival material, long-form takes, and workloads that exceed single-workstation GPU limits.
- Made independent calls on architecture, labelling strategy, post-processing, latency/accuracy trade-offs, and scope reduction under deadline.
- Worked directly with editors and the creative director to define useful model output inside a real production workflow, preserving client data sovereignty on company-controlled infrastructure.

PhD Researcher - Computer Vision and Human Activity Recognition — **Oct 2023 - Oct 2026 expected**

Institute of Robotics and Vision, Auckland University of Technology

- Designed FATL-GCN, a frequency-aware spatio-temporal graph network for skeleton-based human activity recognition; published in Pattern Recognition (Q1, IF 7.6) in 2026.
- First-authored a Computer Science Review survey (Q1 review journal, 2026) on generalisation in skeleton-based HAR, and Dynamic Self-Attention Gated ST-GCN at ICONIP 2024 - owning each from idea and model design through GPU training, ablations and rebuttal.
- Mentor master's students on PyTorch, reproducible experiments, research writing and model evaluation.

Research Assistant - Computer Vision — **Nov 2022 - Jul 2023**

School of Engineering, Computer and Mathematical Sciences, AUT

- Co-built KiwiDetector (improved YOLOv7) and KiwiTracker for real-time orchard video; lead paper has 50+ citations.

- Co-authored three peer-reviewed papers across IVCNZ 2022, SAI Intelligent Systems 2023 and PSIVT 2023.

Teaching Assistant **2022 - present**

Auckland University of Technology

- TA on undergraduate computer-science courses including programming, AI and computer vision; run labs, mark assessments and mentor final-year projects.

Tech Service **Apr - Dec 2023**

PB Tech, Auckland

- Handled hardware diagnostics, software troubleshooting, customer returns and custom PC builds; strengthened plain-English technical communication with non-specialist users.

Junior Lecturer / Co-Founder **2015 - 2021**

Baiyin Vocational and Technology College / Baiyin Pinuo Training School, China

- Taught Computer Fundamentals, Java, IoT applications and RFID technologies to about 120 students per year for four academic years.
- Co-founded a private training school; built curriculum, ran academic operations and finance, then divested before relocating to New Zealand.

EDUCATION

PhD, Computer and Information Sciences **Oct 2023 - Oct 2026 expected**

Auckland University of Technology

AUT PhD Full Scholarship. Thesis: deep learning for skeleton-based human activity recognition.

Master of Computer and Information Sciences **2021 - 2023**

Auckland University of Technology

Specialisation: Computer Vision.

BEng, Logistics Engineering (Software) **2011 - 2015**

Dalian Neusoft University of Information

SELECTED PUBLICATIONS

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). "Beyond Benchmark Accuracy: A Generalization-Centered Analysis of Deep Learning for Skeleton-Based Human Activity Recognition." Computer Science Review, vol. 62. Q1 review journal. doi.org/10.1016/j.cosrev.2026.101030

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). "Frequency-Aware Spatio-Temporal Topology Learning for Skeleton-Based Human Activity Recognition." Pattern Recognition. Q1 journal, IF 7.6. doi.org/10.1016/j.patcog.2026.113146

Y. Xia, M. Nguyen, W. Q. Yan (2022). "A Real-Time Kiwifruit Detection Based on Improved YOLOv7." IVCNZ 2022, Springer LNCS. 50 citations. doi.org/10.1007/978-3-031-25825-1_4

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2025). "Dynamic Self-Attention Gated Spatial-Temporal GCN for Skeleton-Based HAR." ICONIP 2024, Springer LNCS. doi.org/10.1007/978-981-96-6588-4_14. Full publication list on Google Scholar.

AWARDS, SERVICE AND COMMUNICATION

- AUT PhD Full Scholarship - tuition + living stipend, Oct 2023 - Oct 2026.
- SECMS Research Assistant Fund and AUT Summer Research Assistant Scholarship.
- Peer reviewer for ACM TOMM, International Agrophysics and IEEE PIMRC.
- Eight years of cumulative teaching experience across China and New Zealand; bilingual Mandarin Chinese and professional English.

REFEREES

Available on request. Doctoral supervisors at AUT and an industry referee from TOHU MEDIA can be provided.